FlowCrypt: Flow-Based Lightweight Encryption With Near-Lossless Recovery for Cloud Photo Privacy

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***Abstract*—The widespread adoption of smartphone photography has led users to increasingly rely on cloud storage for personal photo archiving and sharing, raising critical privacy concerns. Existing deep learning-based image encryption schemes, typically built upon CNNs or GANs, often depend on traditional cryptographic algorithms and lack inherent architectural reversibility, resulting in limited recovery quality and poor robustness. Invertible neural networks (INNs) have emerged to address this issue by enabling reversible transformations, yet the first INN-based scheme frames encryption as image hiding tasks, relies on an auxiliary reference image and discards by-product information before decryption, leading to degraded recovery and limited practicality. To address these limitations, we propose FlowCrypt, a novel flow-based image encryption framework that simultaneously achieves near-lossless recovery, high security, diversified robustness and lightweight model design. FlowCrypt begins with a secret-key-guided random split to partition input image into complementary representations, enhancing encryption strength without auxiliary references. These components are then fed into a Flow-based Encryption/Decryption (FED) module anchored by conditional affine coupling blocks (CACB) that uniformly confuse dual inputs via key-conditioned affine transformations. Thanks to its reversible architecture and reference-free design, FlowCrypt ensures high-fidelity image recovery. Extensive experiments show that FlowCrypt achieves recovery quality with 115 dB on four datasets, produces uniformly distributed cipher images, exhibits superior resilience to distortions, and maintains a compact architecture with only 1 M parameters, making it suitable for mobile and edge-device applications.**

***Index Terms*—Image encryption, invertible neural network, robust encryption, cloud privacy.**

I. Introduction

With the advancement of high-resolution smartphone cameras, the volume of personal photo data has surged exponentially. Due to limited local storage capacity and the risk of accidental data loss, many users turn to cloud storage services like Google Drive, Apple iCloud for long-term archiving and seamless sharing. However, while cloud storage enhances user convenience, it also exposes private visual content to escalating privacy risks, including unauthorized access, data breaches, and surveillance by third-party providers. As a result, ensuring the confidentiality and integrity of cloud-stored photos has become a pressing challenge in the era of ubiquitous connectivity.

Image encryption offers a viable solution to protect sensitive visual content by transforming it into unintelligible cipher forms, which can only be decrypted by authorized users holding the correct key. However, the demands of real-world cloud storage scenarios impose multiple technical constraints on encryption schemes, including: 1) robustness against statistical and adversarial attacks such as brute-force, differential, and channel-induced noise; 2) lightweight design for real-time execution on resource-constrained edge devices; and 3) high-fidelity decryption to ensure near-pristine recovery of visual quality, which is critical for user satisfaction.

Traditional encryption algorithms such as Data Encryption Standard (DES) [1], International Data Encryption Algorithm (IDEA) [2] and Advanced Encryption Standard (AES) [3], were designed for generic digital data and operate at byte or block levels. Despite their high cryptographic strength, they overlook the spatial redundancies and statistical correlations inherently in images, leading to computational inefficiencies and limited adaptability in image-centric applications. To mitigate these issues, domain-specific encryption schemes have been developed, integrating techniques such as chaotic systems [4], [5], [6], [7], cellular automata (CA) [8], DNA encoding [9], wavelet transforms [10], optical encryption [11] and elliptic curve cryptography [12]. These methods exploit nonlinearity, sensitivity to initial conditions, and permutation-diffusion mechanisms to generate highly diffused and secure ciphers that exhibit superior resistance to differential and statistical attacks. Nevertheless, these approaches remain inherently rule-based and static, limiting their adaptability in dynamic, noise-prone cloud environments.

Recent advances in deep learning have opened up new avenues for image encryption. Deep neural network (DNN)-based methods leverage neural networks' capability for nonlinear transformation and semantic extraction, either to generate keys subsequently integrated into traditional algorithms [13], [14],

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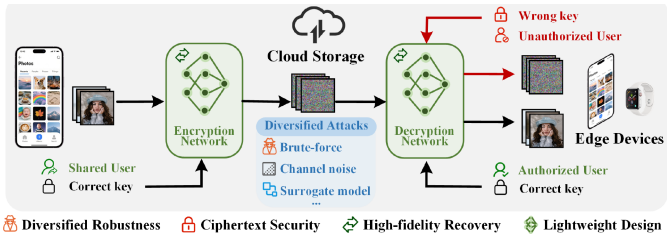

Fig. 1. A paradigm of FlowCrypt for cloud photo protection by invertible image encryption.

[15], [16], or to synthesize encrypted images refined by traditional cryptographic modules [17], [18], [19]. Despite their improved adaptivity, most of these approaches fail to form a truly end-to-end encryption pipeline, often depending on traditional cryptography to fill security gaps. This hybrid design exposes a structural trade-off: purely DNN models are susceptible to statistical attacks due to their lack of formal cryptographic guarantees, whereas reliance on conventional modules struggles to adapt to noisy, non-deterministic cloud conditions.

To address these limitations, invertible neural networks (INNs) have emerged as a promising architecture for reversible image processing tasks, such as image scaling [20], image hiding [21], and digital watermarking [22]. More recently, an INN-based encryption approach [23] integrates encryption and decryption into a unified end-to-end model via bijective mappings. Nevertheless, this framework directly uses the reversible block architecture of HiNet [21], which was originally designed for image hiding. Consequently, it relies on auxiliary reference images and produces discarded by-products, resulting in reduced recovery fidelity and potential security risks when the reference image is exposed.

Motivated by these challenges, we propose FlowCrypt, a novel flow-based image encryption framework that simultaneously achieves near-lossless recovery, diversified robustness, elevated ciphertext security, and a lightweight design. As illustrated in Fig. 1, FlowCrypt enables private photo sharing by transforming input images into noise-like and semantically opaque ciphertexts, thereby protecting sensitive content during cloud transmission and storage under diverse attacks. At the receiver end, authorized users with the correct key can achieve high-fidelity image recovery, whereas unauthorized users can obtain only unrecognizable noise. A secret-key-guided splitting mechanism ensures ciphertext security and key guided recovery, while the encryption-oriented invertible architecture preserves all necessary information for accurate decryption. The shared lightweight encryptor-decryptor module ensures efficient deployment on resource-constrained edge devices with only 1 M parameters. Unlike prior INN-based scheme, FlowCrypt designs a bijective structure that equally confuses two inputs, making it suitable for encryption tasks. Moreover, the secret-key-guided splitting strategy eliminates the need for reference images or discarded intermediary features, thus enhancing recovery fidelity and practical deployability. Ultimately, FlowCrypt constitutes a fully end-to-end image encryption framework, symmetrically coupling encryption and decryption through shared invertible modules. This unified architecture achieves high encryption strength and faithful image reconstruction without reliance on external cryptographic primitives, making it a practical and robust solution for cloud photo privacy in mobile and edge environments. The main contributions of this paper are summarized as follows:

- We design a novel conditional affine coupling block (CACB) for encryption tasks that applies confusion transformations uniformly to both input representations. This is combined with a cipher-oriented regularization scheme that integrates Kullback–Leibler divergence and spatial correlation penalties, promoting uniform distribution and suppressing pixel-wise dependencies in the encrypted output. Unlike prior INN-based schemes derived from flow models for image hiding, where the secret image is transformed to complicate the reference image, CACB uniformly manipulates both inputs conditioned on secret key without relying on multiple transformation rounds or auxiliary randomness. This results in a compact network with only 1 million parameters, delivering superior encryption performance while enabling low-latency encryption and decryption.
- We propose a novel secret-key-guided random splitting strategy that spatially partitions the input image into two complementary representations, aligning with the dual-branch structure of flow models. This strategy injects input diversity and key sensitivity into the encryption process, while ensuring high-fidelity reconstruction during decryption. Crucially, it eliminates the dependence on auxiliary reference images used in prior INN-based methods, thereby enhancing both reconstruction fidelity and practical deployability. To the best of our knowledge, this is the first deep learning-based end-to-end encryption scheme to achieve near-lossless image recovery.
- Comprehensive experiments across multiple datasets along with comprehensive analysis demonstrate that FlowCrypt achieves near-lossless recovery, high ciphertext security, and robust performance under diverse attack conditions with the incorporation of a differentiable noise layer. Additionally, its lightweight design meets the practical requirements of cloud photo privacy scenario, making it suitable for mobile and edge device deployment.

II. Related Work

A. Traditional Image Encryption

Image encryption aims to protect visual content by transforming plain images into noise-like and unintelligible forms through pixel distortion, making it difficult for unauthorized users to decrypt the original content. Early studies [1], [2], [3] treated images as binary data streams, applying conventional cryptographic algorithms originally designed for text. While these ciphers still excel in security and efficiency with modern hardware acceleration like Advanced Encryption Standard New Instructions (AES-NI), their byte- or block-level operations overlook the spatial redundancies and statistical

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correlations inherent in images. This limitation results in suboptimal cipher quality and elevated computational demands for high-resolution visuals, thereby compromising their efficacy in image-specific cloud scenarios. After that, inspired by the pioneering use of chaotic maps in image encryption [24], [25], many encryption schemes start to consider image features, including chaos theory [4], [5], [6], [7], DNA encoding [9], quantum theory [26] and compressive sensing [27]. Among these, chaos theory has emerged as the predominant framework, owing to its chaotic behavior such like ergodic properties and sensitivity to initial states, which conform to the foundational principles of confusion and diffusion in image cryptography. These methods [24], [25], [26], [27], [28], [29], [30] adhere to static, deterministic mathematical rules for pixel manipulation, yielding effective encryption under controlled conditions.

Despite these advances, such rule-based approaches face limitations in dynamic, real-world cloud environments. They struggle to accommodate modern demands such as real-time processing, robustness against diverse attacks, and compatibility with end-to-end learnable systems. These challenges motivate the development of adaptive and data-driven encryption solutions for emerging cloud-based photo storage applications.

B. Deep Learning-Based Image Encryption

In recent years, deep learning (DL)-based image encryption has gained prominence, exploiting the non-linearity and representation capabilities of neural networks to either generate secret keys or directly perform encryption. In key generation schemes, neural networks are employed exclusively for producing secret keys, while subsequent encryption relies on traditional chaotic-based modules. Erkan et al. first combined convolutional neural networks (CNNs) with chaotic maps for image encryption [13]. Ding et al. and Singh et al. further utilized generative adversarial networks (GANs) for key generation, followed by pixel scrambling and diffusion [14], [31]. Related extensions based on DNNs [15], [32], [33] and autoencoders [16] similarly generate chaotic streams for secret key synthesis. However, these approaches treat DL models as mere auxiliary components, with core encryption deferred to traditional encryption schemes, far from achieving end-to-end encryption solutions.

To integrate DNNs more directly into encryption processes, researchers have reframed image encryption as a domain-transfer task, transforming a plain image into a visually inconsistent one [18]. CycleGAN [17] has been employed for medical image encryption, while Zhu et al. employed CNNs for initial encryption and further enhanced security with chaotic systems [19]. Wang et al. designed a DNN-based encryption unit whose network weights were derived from a logistic map [34]. Similarly, Hiroko et al. utilized DNNs for direct encryption, achieving visual indistinguishability but lacking a comprehensive security analysis [35]. Despite these advancements, the continued reliance on chaotic systems limits their end-to-end nature and often suffers from poor robustness. To address these limitations, INNs have been proposed for end-to-end robust encryption [23], employing a shared encryptor-decryptor architecture that enables reversible encryption. However, its reversible block structure is directly adapted from image hiding tasks, requiring the reliance on reference random-noise images for guidance. Additionally, the discarding of intermediary byproducts before decryption limits recovery fidelity and introduces the risk of potential information leakage.

Fig. 2 compares traditional, DNN-based, and INN-based image encryption schemes. Traditional schemes excel in cryptographic security but lack scalability for cloud transmission and storage, where diversified robustness and end-to-end optimization required are essential. DNN-based schemes attempt to combine traditional security strengths with the adaptability of neural networks, but most still rely on traditional modules for security. The first INN-based method achieves end-to-end optimization through invertible architectures. However, it treats encryption in a manner similar to image hiding [21], leading to dependence on auxiliary reference images and discarded intermediate information. These limitations highlight the need for novel encryption schemes that balance end-to-end optimization, high-fidelity recovery, diversified robustness and real-time efficiency for cloud photo storage scenarios.

C. Normalizing Flow-Based Models

Normalizing flow-based models represent a class of invertible neural networks composed of a sequence of bijective mappings that enable invertible transformations between inputs and outputs using the shared parameters. This architecture allows exact recovery of inputs through inverse transformations. Dinh et al. [36] first introduced this paradigm named Non-linear Independent Components Estimation (NICE) to model complex probability densities. Building upon this foundation, RealNVP [37] incorporated convolutional coupling layers to better handle image data, while Glow [38] further enhanced efficiency and photorealistic synthesis by introducing invertible 1×1 convolutions. Subsequent extensions such as i-RevNet [39] integrated residual architectures to improve representational capacity. The inherent reversibility and superior performance of flow-based models make them ideal for tasks requiring precise recovery, including image-to-image translation [40], [41], image scaling [20], super-resolution [42], denoising [43], compression [44], steganography [21], robust watermarking [22], and image encryption [23].

Inspired by HiNet [21], a representative flow-based steganography model renowned for its concealment and reconstruction fidelity, ERIE [23] adapts image hiding architectures by concealing the plain image into a reference noise image through an asymmetric dual-branch design: one branch intricately modulates the input while the other simplifies it. While this asymmetry excels in steganography by selective information suppression, it compromises efficacy in image encryption tasks, as it fails to uniformly diffuse information across representations. To address these limitations, FlowCrypt introduces two pivotal enhancements: symmetric processing of both branches and secret-key-conditioned transformations to amplify key sensitivity. Specifically, as shown in Fig. 3(b) and (d), the affine coupling blocks (ACBs) are symmetrized by applying identical transformations to both the upper and lower branches, thereby

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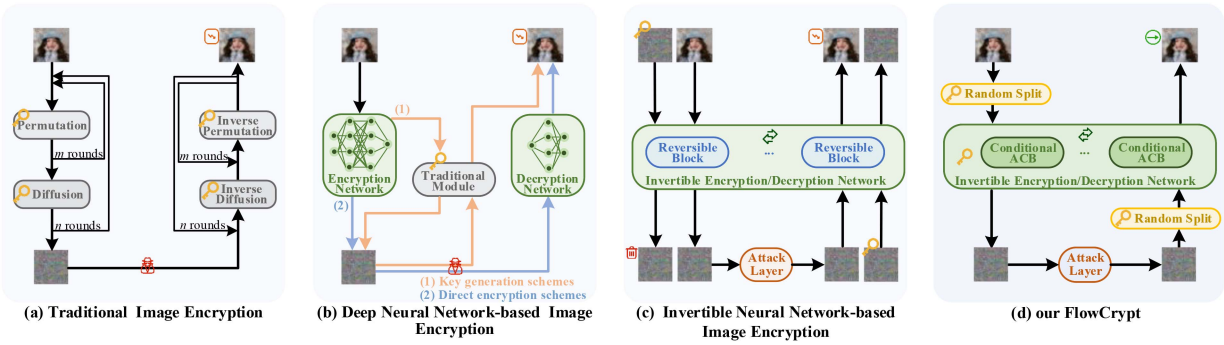

Fig. 2. Evolutionary roadmap of image encryption frameworks. (a) Traditional framework: no end-to-end architecture, high ciphertext security, low diversified robustness and medium recovery fidelity; (b) DNN-based framework: medium end-to-end architecture, medium ciphertext security, low diversified robustness and medium recovery fidelity; (c) first INN-based framework: full end-to-end architecture, high ciphertext security, medium diversified robustness and medium recovery fidelity; (d) our FlowCrypt: full end-to-end architecture, high ciphertext security, high diversified robustness and high recovery fidelity.

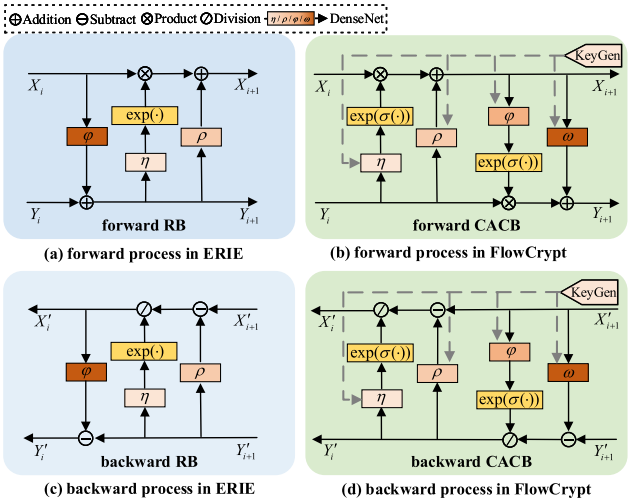

Fig. 3. Architecture comparison of ERIE's reversible block and FlowCrypt's conditional affine coupling block.

ensuring equitable information obfuscation and stronger permutation effects for encryption. Complementing this, we infuse ACBs with key-conditioned affine parameters, enabling key-guided confusion that heightens sensitivity to key perturbations. Furthermore, FlowCrypt integrates a cipher-oriented regularization loss, fusing KL divergence with spatial correlation penalties to enforce uniform histogram distributions and suppress discernible patterns in ciphertexts. Moreover, a novel key-guided splitting mechanism randomly partitions the input image into two complementary representations under key control, seamlessly aligning with the dual-branch ACBs and obviating the need for auxiliary reference images. In contrast to ERIE, which demands multiple reversible transformation rounds and enforced auxiliary randomness to achieve adequate encryption strength, FlowCrypt's conditioned ACBs (CACBs) and regularization demand fewer iterations, yielding a lightweight architecture without sacrificing encryption effect. Coupled with key-guided splitting, this design improves reconstruction fidelity and enhances deployability in resource-constrained cloud storage environments without performance degradation.

III. Threat Model

As an end-to-end image encryption framework, FlowCrypt transforms a plain image I_P into a secure cipher image I_C, enabling near-lossless recovery I'_P for authorized users with the secret key k. This process leverages a key-conditioned random splitting mechanism and a reversible Flow-based Encryption/Decryption (FED) module, formalized as:

$$\begin{aligned} I_C &= f_\theta\left(I_P, G(k)\right) \\ I'_P &= f_\theta^{-1}\left(I_C, G(k)\right) \end{aligned} \tag{1}$$

where f_θ and f_θ^{-1} denote the shared-parameter encryption and decryption functions, respectively, and $G(k)$ injects key-conditioned randomness into the ciphertext generation. The objectives of FlowCrypt encompass: 1) Confidentiality: safeguarding I_C against leakage during cloud storage or insecure transmission, thwarting unauthorized inversion to I_P. 2) Recovery fidelity: enabling authorized recipients with the appropriate secret key k to get high-quality recoveries I'_P.

In cloud storage and sharing scenarios, adversaries $\mathcal{A}$ (insiders like semi-honest cloud providers or outsiders like malicious hackers) attempt to undermine the decryption process beyond legitimate providers and recipients. Their primary objective is to breach the confidentiality of encrypted images I_C, either by intercepting I_C during transmission or inferring the plaintext I_P via legitimate API queries.

The adversary's capabilities are characterized along the following dimensions: 1) Model knowledge: we consider (i) black-box access where $\mathcal{A}$ does not know the trained parameters θ and can only interact with the system through queries; (ii) architecture leakage where $\mathcal{A}$ additionally knows the network architecture/design of the encryption/decryption pipeline but not the trained weights θ. 2) Resource constraints: $\mathcal{A}$ is bounded by a finite query budget and computational resources for training surrogate models. 3) Key knowledge: $\mathcal{A}$ does not have access to the secret key k (nor the key-generation procedure), beyond observing ciphertexts generated under unknown keys. We consider ciphertext-only access (COA), where $\mathcal{A}$ only observes I_C, and chosen-plaintext access (CPA), where $\mathcal{A}$ can query the

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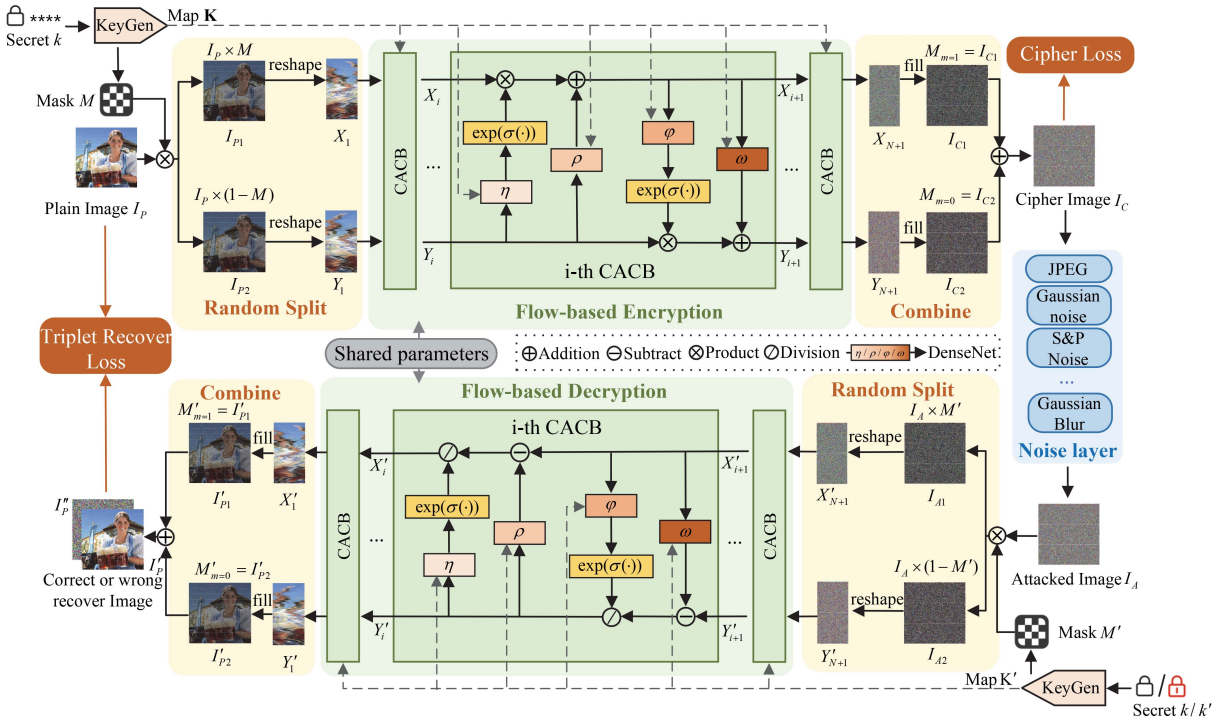

Fig. 4. Overview of the FlowCrypt Framework. The framework comprises three core components: key-guided random split strategy for stochastic input partitioning without auxiliary references, conditional affine coupling blocks (CACBs) for uniform, key-conditioned confusion in flow-based encryption/decryption, and a differentiable noise layer to enhance diversified robustness.

encryption service with chosen I_P to obtain the corresponding I_C, thereby collecting $\left\{\left(I_P^{(i)}, I_C^{(i)}\right)\right\}_{i=1}^{N}$. Given the above assumptions, the adversary's attack process can be formulated as:

$$I_P'' = \mathcal{A}\left(I_C, f_\theta^{-1}\right) \quad \text{or} \quad I_P'' = \mathcal{A}\left(\left\{\left(I_P^{(i)}, I_C^{(i)}\right)\right\}_{i=1}^{N}, f_\theta^{-1}\right) \tag{2}$$

where f_θ^{-1} denotes decryption process parameterized by θ, and $\left\{\left(I_P^{(i)}, I_C^{(i)}\right)\right\}_{i=1}^{N}$ denotes known or chosen plaintext-ciphertext pairs.

In addition to classical cryptanalysis models (ciphertext only, chosen plaintext and differential attacks), we further consider threats powered by deep learning. In a surrogate model attack, $\mathcal{A}$ collects (possibly chosen) (I_P, I_C) pairs via queries and attempts to approximate f_θ^{-1} by training surrogate models [45]. We consider two specific leakage scenarios:

1) *Plaintext-ciphertext pairs leakage:* $\mathcal{A}$ has black-box access to the encryption model, it may submit chosen plaintexts I_P and obtain corresponding ciphertexts I_C. Using a dataset of collected (I_P, I_C) pairs, $\mathcal{A}$ trains a surrogate decryption model with deep leanring models.
2) *Both plaintext-ciphertext pairs and network architecture leakage:* $\mathcal{A}$ additionally knows the encryption/decryption architecture but not the model weights θ. With a dataset of (I_P, I_C) pairs, $\mathcal{A}$ trains a surrogate model with identical model structure. The attack succeeds if the surrogate produces useful reconstructions I_P'' for unseen ciphertexts.

Under the above threat assumptions, commonly used image encryption metrics are interpreted as empirical indicators of resistance to specific attack classes rather than formal cryptographic guarantees: 1) Entropy [46] and pixel correlation measure the statistical unpredictability of ciphertexts, reflecting reduced exploitable structure under COA/CPA-style statistical analysis. 2) NPCR and UACI [47] evaluate the sensitivity of ciphertexts to small perturbations of plaintexts, serving as indicators against differential attacks. The ideal confidence intervals depend on image size, as summarized in Table II. 3) Surrogate reconstruction performance (e.g., PSNR/SSIM of I_P'' w.r.t. I_P under fixed query budgets) directly evaluates robustness against learning-based adversaries with CPA access. Low reconstruction quality indicates that $\mathcal{A}$ fails to recover usable plaintext information from I_C under the assumed threat model. Collectively, these metrics assess whether ciphertexts leak exploitable statistical, differential, or learning-based structure to adversaries operating under realistic deployment assumptions.

IV. Proposed Framework

A. Overview

The overall workflow of FlowCrypt is illustrated in Fig. 4. This framework aims to construct an invertible image encryption framework that simultaneously achieves high security, near-lossless recovery and lightweight model. Given a plain image I_P, a random split strategy conditioned on secret keys is first applied to divide it into two components X_1 and X_2. This step injects randomness and improves the unpredictability of the forward encryption process, thereby enhancing security. The

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resulting components are then passed through the flow-based encryption/decryption (FED) module, which is composed of several invertible neural blocks built upon the ACB structure, well-suited for reversible image encryption and decryption tasks. These blocks further incorporate key-conditioned randomness to strengthen encryption security. The FED module transforms X_1 and X_2 into encrypted representations X_{N+1}, Y_{N+1}, which are then combined to generate the cipher image I_C. To enhance robustness under real-world conditions, a noise simulation layer is introduced to distort the cipher image I_C, producing a degraded image I_A for robust decryption training. The backward decryption takes I_A as input and reconstructs the recovery I'_P via the inverse flow of the FED module. Due to the reversible architecture of FED and less discarded information enabled by the no-reference encryption mechanism, near-lossless recovery can be guaranteed. Furthermore, the FED module shares parameters for both forward encryption and backward decryption, contributing to the model's lightweight design. The detailed design of each key component within FlowCrypt is elaborated as below.

B. Key Generation and Random Split Strategy

The key generation module (KeyGen) is designed to produce pseudorandom, key-independent information that introduces randomness into the FED module. Given a user-defined secret key k, which serves as a password with arbitrary length and characters, a Password-Based Key Derivation Function (PBKDF) is applied to generate a reproducible pseudorandom bitstream with strong key sensitivity. In particular, we adopt PBKDF2 with a high iteration count and a fixed salt to derive a byte stream of length $W \times H$, where W and H are the width and height of the input image I_P. To ensure the generated binary mask is both key-dependent and strictly balanced (i.e., with equal numbers of 0 s and 1 s), we apply a two-step strategy detailed in Algorithm 1. First, we generate a balanced binary vector consisting of exactly half 0 s and half 1 s. Then, the pseudorandom byte stream derived from PBKDF is used to deterministically shuffle this vector presented in Algorithm 2, resulting in a 2D binary mask $M \in \{0,1\}^{W \times H}$, which is used to inject key-independent randomness into the following encryption process:

$$M = \mathcal{T}\left(\text{PBKDF}(k)\right), \quad M \in \{0,1\}^{W \times H} \tag{3}$$

where $\mathcal{T}(.)$ denotes the binarization and balancing transformation applied to the PBKDF output.

The resulting binary mask M is then applied to match each channel of the input image I_O and spatially partition I_O into two complementary parts I_1 and I_2 of the same size:

$$I_{P1} = M \times I_P, \quad I_{P2} = (1 - M) \times I_P \tag{4}$$

where I_{P1} and I_{P2} retain mutually exclusive pixel subsets of I_P, while maintaining the same spatial dimensions.

To prepare the data for invertible encryption, the non-zero pixels in I_{P1} and I_{P2} are reorganized into two tensors X_1 and Y_1 with dimensions $(3, H, W/2)$, using a row-wise and column-wise ordering fashion, respectively. These serve as the paired inputs to the subsequent FED. After the FED processes the inputs and produces the encrypted representations X_{N+1} and Y_{N+1}, the original spatial structure is restored based on a binary mask M. Specifically, a zero-initialized tensor of shape $(H \times W)$ is used to merge the encrypted components: positions corresponding to mask value 1 are filled with X_{N+1}, and those corresponding to 0 are filled with Y_{N+1}, details are presented in Algorithm 3. This strategy ensures that the generated mask is deterministically reproducible from the user-specific password, strictly balanced to preserve the image information symmetry, and highly sensitive to even minor changes in the initial secret key due to the cryptographic properties of PBKDF. Consequently, it enhances the security and unpredictability of the FED module, effectively preventing reverse-engineering of the key or mask from the cipher data. Notably, the PBKDF2 implementation adopted is a Python-based PBKDF2-HMAC-SHA256, build upon the Hash-based Message Authentication (HMAC). Additionally, the secret map K is also generated using the method outlined in Algorithm 1, with the dimension of $(H, W/2)$, matching the size of image features within the ACBs.

Algorithm 1: Binary Mask Generation.

Input: Password k, image dimensions $H \times W$
Output: Mask $M \in \{0,1\}^{H \times W}$
1: $N \leftarrow H \times W$; initialize $S_I \in \{0,1\}^N$ (a "half-1 s, half-0 s" sequence)
2: $k_M \leftarrow \text{PBKDF2}(k, N)$ //use PBKDF2 with password k to derive N bytes of key material k_M
3: $S_R \leftarrow \text{LSB}(k_M) \in \{0,1\}^N$ //Extract LSB sequence
4: $M_S \leftarrow S_I[argsort(S_R + \epsilon)]$ //permute S_I by the order induced by S_R, with a noise ϵ added to break ties
5: $M \leftarrow M_S \in \{0,1\}^{H \times W}$ // reshape M_S into (H, W) to get binary mask M
6: **return** M

Algorithm 2: Random Split.

Input: Password k, plain image $I_P \in \mathbb{R}^{H \times W}$
Output: Split sub-images $X_1, X_2 \in \mathbb{R}^{H \times \frac{W}{2}}$
1: $W, H \leftarrow I_P$ //get shape of image I_P
2: $M \leftarrow \text{BMG}(k, I_P)$ // binary mask generation
3: $X_1 \leftarrow I_P \times M, X_2 \leftarrow I_P \times (1 - M)$ //generate split images
4: reshape X_1 and X_2 to $(H, \frac{W}{2})$
5: **return** X_1, Y_1

C. Flow-Based Encryption/Decryption

Thanks to the strict mathematical reversibility, flow-based network is inherently suitable for image encryption and decryption tasks. The proposed FED consists of two primary components: the forward encryption function f_θ and its inverse f_θ^{-1}, which represents backward decryption. Both functions share the same set of parameters θ. In the forward encryption process, two tensors $X_1, Y_1 \in \mathbb{R}^{3 \times H \times \frac{W}{2}}$ derived from random split module, are sequentially processed by N invertible neural blocks (INBs), yielding two transformed tensors $X_{N+1}, Y_{N+1} \in \mathbb{R}^{3 \times H \times \frac{W}{2}}$, which are then fused to generate the final cipher image. For backward decryption, two input tensors $X'_{N+1}, Y'_{N+1} \in$

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Algorithm 3: Combine.

Input: Encrypted features $X_{N+1}, Y_{N+1} \in \mathbb{R}^{H\times\frac{W}{2}}$, mask $M \in \{0,1\}^{H\times W}$
Output: Cipher image $I_C \in \mathbb{R}^{H\times W}$
1: $W, H \leftarrow X_{N+1}$; initialize $M_0 \in \mathbb{R}^{H\times W}$
2: $I_{C1} \leftarrow M_0[\text{where}(M, X'_{N+1}, 0)]$,
$I_{C2} \leftarrow M_0[\text{where}(\neg M, Y'_{N+1}, 0)]$ //assign X'_{N+1} into M_0 at positions where M is 1 to get I_{C1}, Y'_{N+1} where M is 0 to get I_{C2}
3: $I_C \leftarrow I_{C1} + I_{C2}$ //combine I_{C1}, I_{C2} to get cipher image I_C
4: **return** I_C

$\mathbb{R}^{3\times H\times\frac{W}{2}}$ extracted from the degraded image $I_{\mathcal{A}}$ are passed through the same sequence of INBs in reverse order. This yields the tensors $X'_1, Y'_1 \in \mathbb{R}^{3\times H\times\frac{W}{2}}$ which can further combined to reconstruct the recovered I'_P.

INBs are the core components of the FED framework. As illustrated in Fig. 4, each INB comprises four non-linear mapping functions $\eta()$, $\rho()$, $\varphi()$ and $\omega()$. Each function adopts the same densely connected Conv-LeakyReLu structure, but with independent parameters. Specifically, each mapping uses five 3×3 convolutions (stride 1, padding 1): four 32-channel layers with LeakyReLU and dense concatenation, followed by a linear output convolution. Weights are He-initialized with zero biases, and the last convolution is zero-initialized for stable training. These functions collectively form a CACB, ensuring invertibility. Importantly, each mapping function receives not only the feature maps from plain image, but also secret map $\mathbf{K}$ generated from a secret key k. The secret map $\mathbf{K}$ is concatenated with the image feature maps X_i, Y_i in a channel-wise manner (e.g., replace $\varphi(Y_i)$ with $\varphi(Y_i||\mathbf{K})$ etc). Each nonlinear mapping function maintains the spatial dimensions of the input image feature maps:

$$\eta, \rho, \varphi, \omega : \mathbb{R}^{(C+1)\times H\times\frac{W}{2}} \rightarrow \mathbb{R}^{C\times H\times\frac{W}{2}} \tag{5}$$

where $C = 3$, representing the number of RGB image channels. The transformation within the i^{th} CACB during forward encryption process is formulated as:

$$\begin{aligned} X_{i+1} &= X_i \cdot \exp\left(\sigma(\eta(Y_i||\mathbf{K}))\right) + \rho(Y_i||\mathbf{K}) \\ Y_{i+1} &= Y_i \cdot \exp\left(\sigma(\varphi(X_{i+1}||\mathbf{K}))\right) + \omega(X_{i+1}||\mathbf{K}) \end{aligned} \tag{6}$$

where $i \in [1 \ldots N]$, $||$ denotes channel-wise concatenation and $\sigma(\cdot)$ denotes the sigmoid activation, which served to constrain the image feature map. After being processed by N invertible neural blocks as such, we can obtain X_{N+1} and Y_{N+1} which used to combine the final cipher image.

For backward decryption, the computation proceeds inversely through the CACBs. In particular, for the first CACB, the input data is the split result of the degraded image $I_{\mathcal{A}}$ distorted by the noise layer. For the i^{th} INB, the recovery process between inputs X'_{i+1}, Y'_{i+1} and outputs X'_i, Y'_i is given by:

$$\begin{aligned} Y'_i &= \left(Y'_{i+1} - \omega\left(X'_{i+1}||\mathbf{K}'\right)\right) \cdot \exp\left(-\sigma\left(\varphi\left(X'_{i+1}||\mathbf{K}'\right)\right)\right) \\ X'_i &= \left(X'_{i+1} - \rho\left(Y'_i||\mathbf{K}'\right)\right) \cdot \exp\left(-\sigma\left(\eta\left(Y'_i||\mathbf{K}'\right)\right)\right) \end{aligned} \tag{7}$$

where secret map $\mathbf{K}'$ is regenerated identically using the same secret key k. After passing through all N invertible neural blocks, the final outputs X'_1 and Y'_1 are merged to reconstruct the recovered image. To ensure security, the model is trained such that correct decryption is only possible with the matching key k. When an incorrect key k' is used, the recovered image becomes significantly distorted and visually unrecognizable. In this paper, each non-linear mapping function comprises a stack of 5 "Conv-LeakyReLu" blocks, as illustrated in Fig. 4.

D. Noise Layer

In daily image transmission, the cipher image I_C is often degraded by lossy attacks such as JPEG compression, Gaussian noise, and other distortions. To enhance robustness against such distortions, a differentiable noise layer is introduced between the forward encryption and backward decryption to simulate real-world degradations during training. This layer includes JPEG, Gaussian noise, random crop, Gaussian blur, median blur, cropout, dropout and salt-and-pepper noise. These differentiable operations allow the network to learn to resist common image corruption while maintaining end-to-end trainability.

E. Loss Function

To guarantee the effectiveness of the proposed framework, two different loss functions are introduced: the triplet recovery loss, which ensures lossless image reconstruction, and the cipher loss, which enforces encryption effect and visual randomness.

1) Triplet Recovery Loss: The aim of the backward decryption process is to accurately recover the plain image. To achieve this, Mean Squared Error (MSE) loss is applied to measure the discrepancy between images. To further enhance robustness, especially under incorrect key scenarios, a triplet loss is introduced to contrastly optimize correct and incorrect reconstructions. The full triplet recovery loss is defined as:

$$\begin{aligned} \mathcal{L}_{TriRecover}(\theta) &= \mathcal{L}_{TriMSE}(A, P, N) \\ &= \mathcal{L}_{TriMSE}(I_P, I'_P, I''_P) \\ &= \max\left\{0, MSE\left(I_P, I'_P\right) - MSE\left(I_P, I''_P\right) + 1.0\right\} \end{aligned} \tag{8}$$

where $\mathcal{L}_{TriMSE}$ denotes a triplet loss based on MSE distance metric, $A = I_P$ (anchor: the plain image), $P = I'_P$ (positive: the correctly decrypted image), and $N = I''_P$ (negative: the wrong decrypted image using a wrong key). The triplet loss encourages the correct reconstruction to be closer to the plain image while push away the incorrect reconstruction, even in the absence of direct supervision for the latter.

2) Cipher Loss: To ensure the security and visual unpredictability of the cipher image I_C in the forward encryption process, a cipher loss is introduced, consisting of two components:

$$\mathcal{L}_{Cipher}(\theta) = \mathcal{L}_{uniform} + \mathcal{L}_{corr} \tag{9}$$

where $\mathcal{L}_{uniform}$ represents uniform distribution loss which encourages the histogram distribution of I_C to approximate a uniform distribution, thus reducing statistical redundancy:

$$\mathcal{L}_{uniform} = D_{KL}(P(h_{I_C})||U) \tag{10}$$

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where D_{KL} denotes the Kullback-Leibler divergence, which is used to measure the similarity of two distributions, $P(h_{I_C})$ is the estimated histogram probability of I_C over 256 gray levels, and $U = \frac{1}{256}$ denotes the target uniform distribution. Additionally, $\mathcal{L}_{corr}$ denotes pixel correlation loss which minimizes spatial correlation between adjacent pixels in I_C to enhance visual randomness:

$$\mathcal{L}_{corr} = |\rho_{xy}| = \left| \frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{N}(x_i - \bar{x})^2 \times \sum_{i=1}^{N}(y_i - \bar{y})^2}} \right| \tag{11}$$

where x_i and y_i denote values of adjacent pixels in I_C, N is the total number of the sampled pixel pairs, and $\bar{x} = 1/N \left(\sum_{i=1}^{N}\right) x_i, \bar{y} = 1/N \left(\sum_{i=1}^{N}\right) y_i$ are their respective means.

3) Total Loss: The total loss function $\mathcal{L}_{Total}$ is a weighted sum of cipher loss $\mathcal{L}_{Cipher}$ and triplet recovery loss $\mathcal{L}_{TriRecover}$:

$$\mathcal{L}_{Total}(\theta) = \lambda_1 \mathcal{L}_{Cipher} + \lambda_2 \mathcal{L}_{TriRecover} \tag{12}$$

where λ_1 and λ_2 are weight factors to balance encryption strength and reconstruction fidelity.

V. Experimental Results

A. Experimental Settings

1) Datasets and Settings: The DIV2K [48] training dataset (800 images) is used for training FlowCrypt, with training images resized to 112×112. For evaluation, we use four datasets: the DIV2K testing set (100 images), COCO [49] (5,000 images), ImageNet [50] (50,000 images) and CelebA-HQ (30,000 images). CelebA-HQ consists of large-scale real-world portrait images, which are representative of personal photos commonly stored in cloud-based photo services. All test images are resized to 256×256. The number of ACBs is set as 4 as it achieves the trade-off between performance and model complexity. The parameters of λ_1 and λ_2 are fixed as 5.0 and 5.0, respectively. The entire training process converges after approximately 300000 steps (corresponding to 4000 epochs), taking around 30 hours on a single NVIDIA RTX 2080Ti GPU. The training batch size is set to 10. Under this configuration, each training iteration requires approximately 0.45 seconds. The framework is implemented by PyTorch [51]. Optimization is performed using Adam [52] with $\beta_1 = 0.9$ and $\beta_2 = 0.999$, and an initial learning rate of $1 \times 10^{-4.5}$.

2) Benchmarks: To verify the recovery quality and encryption security of proposed method, we compare it with several state-of-the-art (SOTA) image encryption methods, including the first INN-based end-to-end robust image encryption (ERIE) method [23], CNN-based method CNN-IE [13] and a traditional chaotic encryption scheme LTMM-CIEA [28]. For a fair comparison, all methods are evaluated on images resized to 256×256.

3) Evaluations: To measure the quality of recovered images, we employ peak signal-to-noise ratio (PSNR), structural similarity index (SSIM) [53] to evaluate the recovery quality, covering various scenarios including correct recoveries, incorrect recoveries with wrong keys, and recoveries against distortions or attacks. Additionally, root mean square error (RMSE), and mean absolute error (MAE) are computed to further assess the pixel-wise statistical similarity between the recovered image and the original plain image. Higher PSNR and SSIM, along with lower RMSE and MAE, indicate better recovery quality. Conversely, for incorrect recoveries, lower PSNR and SSIM values reflect higher resistance to unauthorized access, and are therefore preferred.

To evaluate the encryption effect of cipher images, we use histogram analysis, information entropy and pixel correlation in horizontal, vertical and diagonal directions. These metrics respectively measure the pixel distribution, the degree of randomness and the correlation between adjacent pixels. A more uniform histogram, an entropy value closer to the ideal value of 8, and lower correlation values indicate better encryption performance. The entropy is calculated as:

$$H(I) = \sum_{i=1}^{F} Pr(x_i) log_2 Pr(x_i) \tag{13}$$

where F is the total number of possible pixel numbers, x_i denotes the i-th pixel value, and $Pr(x_i)$ represents its probability. When each possible pixel value has equal probability, the image achieves its theoretical maximum information entropy. For an 8-bit image, this is $H(I) = log_2 F = log_2 2^8 = 8$. The correlation coefficient C between two neighboring pixels x and y can be computed as:

$$C(x, y) = \frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{N}(x_i - \bar{x})^2 \sum_{i=1}^{N}(y_i - \bar{y})^2}} \tag{14}$$

where N is the total number of pixel pairs considered, x_i, y_i are the pixel values of two neighboring pixels, $\bar{x}, \bar{y}$ are the mean pixel values of considered pixel pairs. When the pixels are uncorrelated, the correlation coefficient approaches 0, indicating a high level of randomness and a more secure encryption.

To quantitatively evaluate the key sensitivity, we employ NPCR and UACI metrics. These metrics measure how significantly the cipher image changes when a small variation is introduced in the encryption key or plaintext. For an image with 8-bit pixels, the theoretical ideal values are approximately NPCR $= 99.61\%$ and UACI $= 33.46\%$. Values of NPCR and UACI that are closer to the ideal values indicate stronger key sensitivity and better robustness against differential attacks. The formulas for NPCR and UACI are given as follows:

$$\text{NPCR} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \frac{B(i,j)}{255} \tag{15}$$

$$B(i,j) = \begin{cases} 0, if\ C_1(i,j) = C_2(i,j) \\ 1, if\ C_1(i,j) \neq C_2(i,j) \end{cases} \tag{16}$$

$$\text{UACI} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \frac{|C_1(i,j) - C_2(i,j)|}{255} \tag{17}$$

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TABLE I
BENCHMARK COMPARISONS OF RECOVERY PERFORMANCE

Method	Image size	DIV2K PSNR(dB)↑	DIV2K SSIM↑	DIV2K MAE↓	DIV2K RMSE↓	COCO PSNR(dB)↑	COCO SSIM↑	COCO MAE↓	COCO RMSE↓
ERIE	128 × 128	50.87±0.61	0.9989±2.72e-04	1.81e-03±7.06e-05	2.86e-03±1.92e-04	50.88±0.58	0.9988±2.22e-04	1.79e-03±6.82e-05	2.86e-03±1.97e-04
	256 × 256	51.81±0.68	0.9986±3.98e-04	1.67e-03±7.42e-05	2.57e-03±1.90e-04	51.98±0.48	0.9986±2.48e-04	1.64e-03±5.20e-05	2.52e-03±1.42e-04
	512 × 512	52.50±0.71	0.9984±4.20e-03	1.59e-03±7.18e-05	2.37e-03±1.89e-04	52.63±0.43	0.9984±2.65e-04	1.55e-03±4.60e-05	2.33e-03±1.18e-04
FlowCrypt	128 × 128	115.86±0.13	0.9999±2.84e-06	1.06e-06±1.78e-08	1.61e-06±2.45e-08	115.83±0.14	0.9999±2.03e-06	1.06e-06±1.87e-08	1.61e-06±2.62e-08
	256 × 256	115.78±0.09	0.9999±3.15e-06	1.08e-06±1.34e-08	1.62e-06±1.72e-08	115.76±0.09	0.9999±2.36e-06	1.08e-06±1.29e-08	1.62e-06±1.76e-08
	512 × 512	115.71±0.08	0.9999±3.20e-06	1.09e-06±1.24e-08	1.63e-06±1.56e-08	115.69±0.07	0.9999±2.55e-06	1.09e-06±1.10e-08	1.64e-06±1.46e-08
Method	Image size	ImageNet PSNR(dB)↑	ImageNet SSIM↑	ImageNet MAE↓	ImageNet RMSE↓	CelebA-HQ PSNR(dB)↑	CelebA-HQ SSIM↑	CelebA-HQ MAE↓	CelebA-HQ RMSE↓
ERIE	128 × 128	50.82±1.01	0.9987±8.49e-04	1.78e-03±1.29e-04	2.89e-03±3.54e-04	51.85±0.39	0.9987±2.01e-04	1.68e-03±4.23e-05	2.55e-03±1.17e-04
	256 × 256	51.86±0.92	0.9984±8.74e-04	1.64e-03±1.18e-04	2.56e-03±2.89e-04	53.01±0.31	0.9983±1.90e-04	1.53e-03±3.51e-05	2.23e-03±8.24e-05
	512 × 512	52.61±0.90	0.9982±8.67e-04	1.54e-03±1.09e-04	2.35e-03±2.64e-04	53.74±0.25	0.9981±2.40e-04	1.45e-03±2.84e-05	2.05e-03±6.02e-05
FlowCrypt	128 × 128	115.82±0.14	0.9999±3.78e-06	1.06e-06±1.91e-08	1.61e-06±2.65e-08	115.77±0.13	0.9999±1.85e-06	1.07e-06±1.83e-08	1.62e-06±2.48e-08
	256 × 256	115.76±0.09	0.9999±4.63e-06	1.08e-06±1.38e-08	1.62e-06±1.82e-08	115.71±0.07	0.9999±2.40e-06	1.08e-06±1.14e-08	1.63e-06±1.47e-08
	512 × 512	115.69±0.08	0.9999±5.10e-06	1.09e-06±1.21e-08	1.64e-06±1.56e-08	115.66±0.05	0.9999±2.77e-06	1.10e-06±9.22e-09	1.64e-06±1.08e-08

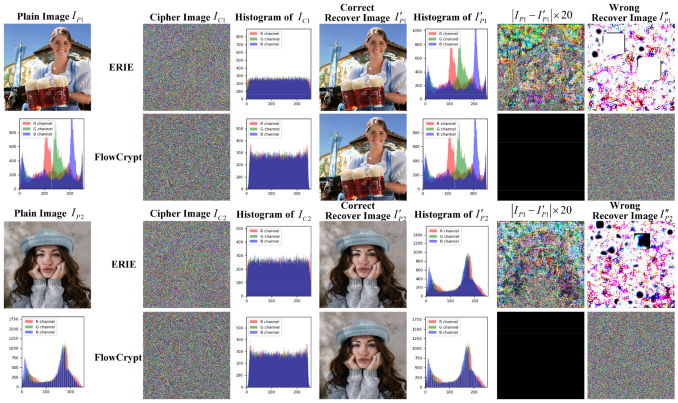

Fig. 5. Visual comparison results of FlowCrypt and ERIE.

B. Encryption and Decryption Effect

1) Qualitative Results: As aforementioned, the proposed FlowCrypt could achieve high-fidelity image recovery due to the inherent invertibility of the flow-based framework. As illustrated in Fig. 5, qualitative visual comparison results underscore its superior encryption effect and near-lossless reconstruction capabilities. The encrypted cipher images appear as noise-like patterns with no discernible visual structures, and their corresponding histograms exhibit uniform distributions that effectively conceal semantic information. The correctly recovered images closely resemble the original plain images, while wrong decryptions yield markedly distorted outputs. Moreover, the residual images (magnified 20 times) between the correct recoveries and ground-truth plain images appear uniformly black, signifying negligible reconstruction errors. By contrast, residuals from ERIE exhibit noticeable artifacts, reflecting inferior recovery performance. These results confirm that FlowCrypt achieves both secure encryption effect and near-lossless reconstruction fidelity.

2) Quantitative Results: Table I presents the quantitative comparison between FlowCrypt and ERIE. Each cell presents the mean value along with the corresponding standard deviation. As shown, FlowCrypt outperforms ERIE across all four metrics. Specifically, it achieves mean PSNR gains of 64.05 dB, 63.93 dB, 63.99 dB, 62.84 dB over the INN-based baseline ERIE on the DIV2K, COCO, ImageNet and CelebA-HQ datasets, respectively, across image resolutions of 128×128, 256×256 and 512×512 pixels. Similar improvements are observed in SSIM, RMSE, and MAE, collectively highlighting the superior reconstruction quality of FlowCrypt. These improvements stem from the reversible nature of the flow-based architecture, and preserves all information throughout the encryption-decryption process, enabling near-lossless recovery. Notably, although our model is trained solely on the DIV2K dataset, it generalizes well to COCO, ImageNet and CelebA which reflects real-world personal photo characteristics, demonstrating robust cross-domain performance, which is critical for practical applications such like cloud photo storage.

C. Security Analysis

1) Key Security: For key space, a larger key space enhances resistance to brute-force attacks. In FlowCrypt, the key consists of two components: 1) a random mask M used to split the plain image, with size of 2^{W*H}; and 2) a secret map $\mathbf{K}$ that interacts with image features in the ACBs, with size of $2^{W*H/2}$. Both the random mask M and secret map $\mathbf{K}$ are deterministically derived from the user's password using PBKDF2. Therefore, the total key space is governed by the password entropy and PBKDF2 parameters, and can be expressed as C^L, where C represents the size of the character set and L is the length of the user's key. For instance, consider a commonly used password configuration: a 16-character key chosen from the ASCII character set, which includes 95 printable characters. In this case, the total key space is $95^{16} \approx 10^{31.64} \approx 2^{105.12}$. This expansive key space is sufficient to resist exhaustive brute-force attacks.

In terms of key sensitivity, a secure encryption system should exhibit high sensitivity to key variations, which means that even a single-bit change in the secret key should lead to significant differences in the ciphertexts and recovered images. As shown in Fig. 6, two cipher images I_{C1} and I_{C2}, encrypted from the same plaintext using two keys differing by 1-bit, exhibit stark disparities, with their residual revealing complete difference. During decryption, the correct key yields precise recovery I'_{P1}, whereas a 1-bit change in secret key k results in a completely

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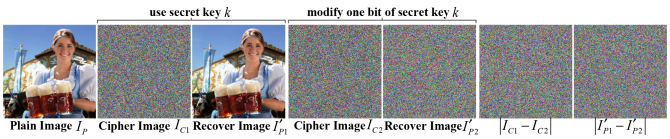

Fig. 6. Examples of encryption and decryption results under different keys with slight changes.

TABLE II
BENCHMARK COMPARISONS OF KEY SECURITY AND CIPHERTEXT SECURITY ANALYSIS

Method	NPCR (99.61%)	UACI (33.46%)	Entropy (8)
CNN-IE	98.82	29.11	7.9989
ERIE	99.58	32.94	7.9968
FlowCrypt	99.38	32.52	7.9979
Method	**Local entropy (R)**	**Local entropy (G)**	**Local entropy (B)**
CNN-IE	7.9025±0.0018/39.1	7.9024±0.0018/38.4	7.9024±0.0019/40.2
ERIE	7.9019±0.0049/16.2	7.9021±0.0051/13.9	7.9019±0.0050/15.2
FlowCrypt	7.9021±0.0052/17.0	7.9021±0.0050/15.8	7.9019±0.0050/15.6
Method	**Correlation (H)↓**	**Correlation (V)↓**	**Correlation (D)↓**
CNN-IE	0.0011	-0.0003	0.0034
ERIE	0.0002	0.0038	0.0014
FlowCrypt	0.0029	0.0001	-0.0023

Note: For local entropy, the value after "/" denotes the pass rate (%) of the corresponding local randomness test under $\alpha = 0.001$.

distorted recovery I'_{P2}, demonstrating the effectiveness of key sensitivity. To quantitatively evaluate key sensitivity, we compute NPCR and UACI between ciphertext pairs generated from the same plaintext under two keys that differ by one bit. The results on 1000 randomly selected COCO images are reported in Table II. FlowCrypt achieves NPCR and UACI values that are close to the expected values for 8-bit images ($\text{NPCR}_{Expected} = 99.61\%$, $\text{UACI}_{Expected} = 33.46\%$), further confirming its strong key sensitivity.

2) Ciphertext Security: As for ciphertext security, we analyze the information entropy, pixel correlation and National Institute of Standards and Technology (NIST) Statistical Test Suite (STS) [54] for cipher images. Table II lists the corresponding results over 1000 cipher images. FlowCrypt achieves entropy values close to the theoretical ideal of 8, indicating uniformly global pixel distributions. Since global entropy alone may overlook local non-uniform patterns, we further evaluate the local Shannon entropy (LSE) following Wu et al. [46]. For each 256×256 color cipher image, the (K, T_B)-LSE is computed as the average Shannon entropy over $K = 30$ randomly selected non-overlapping blocks, each containing $T_B = 1936$ pixels. Under $\alpha = 0.001$, the acceptance interval for an ideally random 8-bit single-channel image is $[7.901515698,\ 7.903422936]$ [46]. As reported in Table II, FlowCrypt achieves channel-wise local entropy values of 7.9021, 7.9021 and 7.9019, with corresponding pass rates of 17.0%, 15.8% and 15.6%. Compared with the INN-based scheme ERIE, FlowCrypt yields consistently higher pass rates across all three channels, indicating stronger local randomness. However, it still underperforms CNN-IE, possibly because FlowCrypt does not explicitly constrain local block randomness in its training loss, whereas CNN-IE employs conventional pixel permutation and diffusion operations

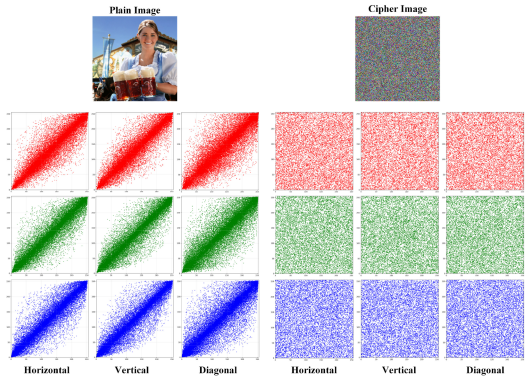

Fig. 7. Distribution of adjacent pixels in different directions for plain image and cipher image. Three rows represent the visualization results on the R, G, and B channels, respectively.

TABLE III
NIST SP800-22 TEST RESULTS FOR THE CIPHER IMAGES GENERATED BY FLOWCRYPT AND ERIE

Statistical test	ERIE			FlowCrypt		
	P-value$_T$	Prop.[a]	Result	P-value$_T$	Prop.[a]	Result
Frequency	0.897763	1.00	Pass	0.071177	0.99	Pass
Block frequency(M=128)	0.030806	0.99	Pass	0.657933	1.00	Pass
Runs	0.616305	0.99	Pass	0.816537	0.98	Pass
Longest runs of ones	0.534146	1.00	Pass	0.816537	1.00	Pass
Rank	0.996335	0.98	Pass	0.514124	1.00	Pass
FFT	0.289667	1.00	Pass	0.010237	0.99	Pass
Non-overlapping templates(m=9)[b]	0.171867	0.95	Fail	0.171867	0.97	Pass
Overlapping templates(m=9)	0.224821	1.00	Pass	0.955835	0.98	Pass
Universal	0.437274	1.00	Pass	0.978072	0.97	Pass
Linear complexity(M=500)	0.851383	1.00	Pass	0.534146	0.99	Pass
Serial(m=16) P-value1	0.213309	1.00	Pass	0.851383	1.00	Pass
Serial(m=16) P-value2	0.455937	1.00	Pass	0.289667	1.00	Pass
Approximate entropy(m=10)	0.013569	1.00	Pass	0.437274	0.99	Pass
Cumulative sums (Forward)	0.383827	1.00	Pass	0.080519	0.97	Pass
Cumulative sums (Reverse)	0.474986	1.00	Pass	0.657933	0.99	Pass
Random excursions[b]	0.026648	0.98	Pass	0.095617	0.97	Pass
Random excursions Variant[b]	0.141256	0.97	Pass	0.086458	0.98	Pass

[a] The proportion of sequences that passed the NIST randomness test.
[b] The minimum proportion obtained from multiple test.

that are more favorable for improving local randomness. Additionally, the average correlations in horizontal, vertical, and diagonal directions are all close to zero, suggesting minimal correlation between adjacent pixels and strong resistance to statistical analysis. These attributes are shown in Fig. 7, where adjacent pixels in plain images exhibit clustered distributions while diffused, uncorrelated scatters in cipher images. To further evaluate the statistical randomness of the generated cipher images, NIST SP800-22 [54] is employed. A $256 \times 256 \times 3$ image from COCO dataset is encrypted 100 times using different random secret keys, and each encrypted image is converted into a binary sequence of length $256 \times 256 \times 3 \times 8 = 1572864$ bits. The significance level is set to $\alpha = 0.01$ as recommended by NIST. For each sub-test, Table III reports corresponding P-value and passing proportion (Prop.) over 100 rounds. Following the NIST criterion [54], the minimum acceptable passing proportion is $T = (1-\alpha) - 3\sqrt{\frac{\alpha(1-\alpha)}{m}}$, where m is the number of tested sequences. For $m = 100$, this gives $T \approx 0.96015$, meaning that at least 97 sequences must pass each sub-test. Moreover, the P-value distribution is regarded as uniform when

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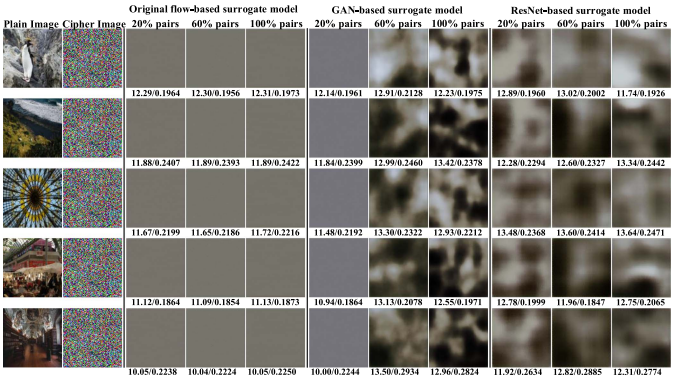

Fig. 8. Examples of decryption results against surrogate attacks. The data under decryption results represent the PSNR and SSIM of decrypted image with respect to plain image.

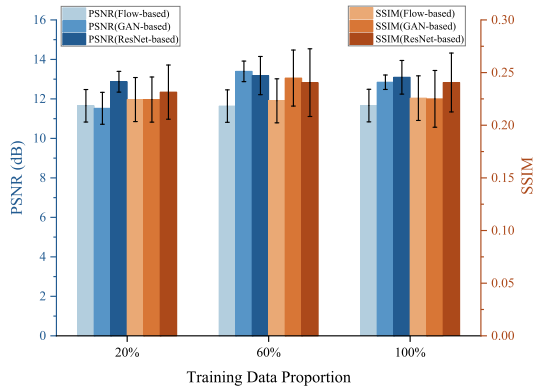

Fig. 9. Quantitative results of FlowCrypt against surrogate attacks.

P-value$_T \geq 0.0001$. As reported in Table III, FlowCrypt passes all the reported sub-tests, indicating that its ciphertext sequences exhibit strong statistical randomness under the NIST criteria. While ERIE fails the non-overlapping template test, revealing the existence of weak residual local patterns in its ciphertext bitstreams. Combined with the entropy, local entropy, NPCR, UACI, correlation and NIST analyses, FlowCrypt exhibits strong statistical security and effectively prevents adversaries from inferring plaintext semantics from ciphertexts.

3) Surrogate Model Attack: To access whether attackers can train effective surrogate deceyption models, we consider two leakage representative scenarios: 1) leakage of plaintext-ciphertext pairs; and 2) simultaneous leakage of encryption outputs and the network architecture. To comprehensively evaluate the resistance capacity against such surrogate model attacks of FlowCrypt, we trained GAN-based model and ResNet-based model for scenario 1), and an original flow-based model for scenario 2). Plaintexts are sampled from DIV2K dataset, with training data proportions of 20%, 60% to 100%, and all models are trained for 100 epochs. As shown in Fig. 8, across both leakage scenarios, the decryption results from all surrogate models, including original INN-based, GAN-based and ResNet-based surrogate decryption models, exhibit indistinguishable patterns. Quantitative results in Fig. 9 further confirm this: PSNR values

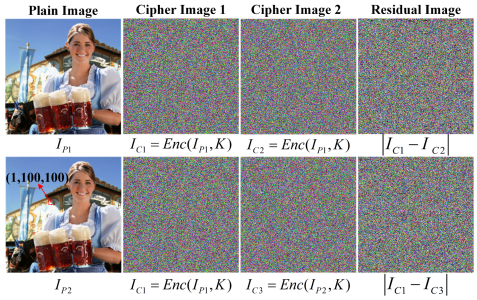

Fig. 10. Examples of FlowCrypt to resist chosen plaintext attack and differential attack. Plain image I_{P2} is a one-bit difference compared with I_{P1} at position (1,100,100).

remain below 15 dB and SSIM below 0.3, even as the proportion of training data increases from 20% to 100%. These results underscore FlowCrypt's robust defense against surrogate model and imitation-learning attacks, irrespective of partial data or structural exposures.

4) Traditional Attack Models: In ciphertext-only attacks (COA), the attacker only has access to ciphertext and attempts to recover either the plaintext or the secret key. The key space of FlowCrypt is approximately 2^{105}, derived from a user-defined 16-character password, which is large enough to resist brute-force attacks. Additionally, the statistical properties of cipher images exhibit high information entropy, uniform histogram distribution and uncorrelated adjacent pixels, which collectively demonstrate strong resilience to statistical cryptanalysis under COAs.

In chosen-plaintext attacks (CPA) and differential attacks, the attacker can actively select plaintexts and obtain corresponding ciphertexts, making CPA more powerful than known-plaintext attacks. FlowCrypt defends against CPA by introducing key-conditioned randomness into each ACB, thereby enhancing diffusion throughout the encryption process. Fig. 10 illustrates FlowCrypt's ability to resist CPA. In the first row, the same plaintext encrypted twice under the same secret key produces completely different ciphertexts. In the second row, changing only a single bit in the plaintext leads to a substantial difference in the resulting ciphertext. These visual results demonstrate FlowCrypt's robustness against both chosen-plaintext and differential attacks. To further quantify resistance to differential attacks, Table IV reports NPCR and UACI results under the size-dependent statistical criteria in [47] with significance level $\alpha = 0.001$. The corresponding criteria are as follows: for 128×128, NPCR $\geq 99.4588\%$ and UACI $\in [32.8552\%, 34.0718\%]$; for 256×256, NPCR $\geq 99.5341\%$ and UACI $\in [33.1594\%, 33.7677\%]$; for 512×512, NPCR $\geq 99.5717\%$ and UACI $\in [33.3115\%, 33.6156\%]$; and for 1024×1024, NPCR $\geq 99.5906\%$ and UACI $\in [33.3875\%, 33.5396\%]$. For this evaluation, 12 color images are randomly selected from COCO dataset, with three images for each resolution. For each image, a one-bit perturbation is introduced into the plaintext, and the resulting ciphertext pair is used to compute NPCR and UACI separately on the R, G, and B channels. In the comparative study, we focus on FlowCrypt and ERIE, while CNN-IE is

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TABLE IV
NPCR AND UACI RESULTS OF FLOWCRYPT AND ERIE ON COLOR IMAGES OF DIFFERENT SIZES ($\alpha = 0.001$)

Image size	File name	NPCR (%)				UACI (%)			
		ERIE (R/G/B)	Result	FlowCrypt (R/G/B)	Result	ERIE (R/G/B)	Result	FlowCrypt (R/G/B)	Result
128×128×3	000000048555	99.5727/99.6215/99.6276	Pass	99.4812/99.5178/99.5117	Pass	33.3070/33.0066/33.2877	Pass	33.0363/33.1523/33.2897	pass
	000000064898	99.5422/99.7070/99.6948	Pass	99.5544/99.5239/99.5849	Pass	33.3993/33.6550/33.2862	Pass	33.1284/33.3341/33.5338	pass
	000000071756	99.6337/99.6459/99.6215	Pass	99.5422/99.5178/99.5056	Pass	33.3257/33.3750/33.1639	Pass	33.3122/33.2150/33.4346	pass
256×256×3	000000034139	99.6414/99.6109/99.6383	Pass	99.5574/99.5529/99.5941	Pass	**33.0713**/33.2487/33.2187	Fail	33.4264/33.3805/33.4052	Pass
	000000042296	99.6093/99.6292/99.6063	Pass	99.5452/99.6200/99.5483	Pass	33.4697/33.2819/33.2366	Pass	33.5918/33.4752/33.3809	Pass
	000000049269	99.6398/99.6139/99.6124	Pass	99.5727/99.5941/99.5437	Pass	33.3982/33.3427/33.2297	Pass	33.4327/33.3470/33.3687	Pass
512×512×3	000000042178	99.6047/99.6105/99.6124	Pass	99.5742/99.5887/99.5754	Pass	33.3772/33.3301/**33.1701**	Fail	33.3545/33.4231/33.3871	Pass
	000000115946	99.6089/99.6139/99.6006	Pass	99.5868/99.6017/99.6070	Pass	**33.3076**/33.3224/**33.2844**	Fail	33.3566/33.4452/33.3523	Pass
	000000492282	99.5861/99.5857/99.6120	Pass	99.6276/99.6051/99.6120	Pass	**33.2937**/33.3786/**33.2085**	Fail	33.3270/33.3600/33.3245	Pass
1024×1024×3	000000010977	99.6154/99.5997/99.6126	Pass	99.6043/99.5972/99.5992	Pass	**33.2930/33.2937/33.2637**	Fail	33.3396/33.4830/33.3933	Pass
	000000018380	99.6132/99.6048/99.6157	Pass	99.5960/99.5940/99.6018	Pass	**33.2848/33.2686/33.2759**	Fail	33.3950/33.4676/33.3876	Pass
	000000088269	99.6103/99.6068/99.6065	Pass	**99.5882**/99.5915/99.5943	Fail	33.3129/**33.2804/33.2379**	Fail	33.3567/33.4985/33.3652	Pass

Note: Bold values indicate the channel(s) that fail the corresponding size-matched criteria.

TABLE V
BENCHMARK COMPARISONS OF RECOVERY PERFORMANCE AGAINST COMMON DISTORTIONS (PSNR/SSIM IN EACH CELL)

Method	Cropout (%)					Dropout (%)				
	r=0.1	0.2	0.3	0.4	0.5	r=0.2	0.3	0.4	0.5	0.6
LTMM-CIEA	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.007
CNN-IE	27.94/0.916	22.12/0.745	18.55/0.574	16.09/0.440	14.11/0.331	15.10/0.380	13.34/0.283	12.09/0.216	11.12/0.164	10.33/0.123
ERIE_p	6.17/0.481	5.42/0.403	4.85/0.339	4.49/0.301	4.37/0.296	4.41/0.308	4.41/0.308	4.41/0.308	4.41/0.308	4.41/0.308
ERIE_n	27.49/0.965	20.74/0.925	17.14/0.865	14.48/0.785	12.30/0.679	6.15/0.014	6.15/0.014	6.15/0.014	6.16/0.015	6.16/0.015
FlowCrypt_p	9.53/0.112	10.64/0.108	10.76/0.101	10.85/0.089	10.84/0.081	10.74/0.343	10.81/0.325	10.79/0.303	10.76/0.242	10.74/0.242
FlowCrypt_n	**31.74/0.971**	**25.54/0.936**	**22.10/0.891**	**19.52/0.830**	**17.49/0.755**	**19.82/0.573**	**18.22/0.496**	**17.10/0.439**	**16.25/0.392**	**15.49/0.353**
Method	SP Noise (%)					JPEG Compression (%)				
	r=0.01	0.02	0.03	0.04	0.05	Q=50	60	70	80	90
LTMM-CIEA	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.007
CNN-IE	28.12/0.886	25.10/0.804	23.33/0.742	22.08/0.692	21.12/0.651	11.53/0.104	11.92/0.116	12.28/0.128	12.61/0.140	13.06/0.157
ERIE_p	6.55/0.091	5.14/0.158	4.66/0.240	4.50/0.282	4.45/0.294	4.93/0.023	4.88/0.034	4.78/0.088	4.65/0.176	4.54/0.249
ERIE_n	16.60/0.473	13.96/0.304	12.61/0.241	11.61/0.205	10.80/0.175	14.60/0.278	15.51/0.337	16.25/0.417	16.44/0.513	16.59/0.622
FlowCrypt_p	11.40/0.325	11.34/0.302	11.35/0.279	11.36/0.257	11.35/0.239	10.98/0.088	10.90/0.095	10.90/0.082	10.84/0.078	10.67/0.077
FlowCrypt_n	**29.46/0.888**	**26.66/0.811**	**25.10/0.750**	**23.95/0.699**	**23.09/0.656**	**30.01/0.893**	**28.76/0.871**	**27.89/0.856**	**27.29/0.844**	**26.89/0.834**
Method	Gaussian Noise (%)			Gaussian Blur (%)			Median Blur (%)			
	$\sigma = 0.01$	0.03	0.05	$\sigma = 0.5$	1	2	$w = 3$	5	7	9
LTMM-CIEA	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.007	7.97/0.008	7.97/0.007	7.97/0.008	7.97/0.008
CNN-IE	11.83/0.195	10.08/0.110	9.49/0.081	9.77/0.093	7.67/0.032	7.87/0.011	8.33/0.011	8.14/0.007	8.12/0.007	8.12/0.007
ERIE_p	6.36/0.019	5.41/0.008	5.20/0.006	5.03/0.003	4.89/0.033	4.84/0.067	4.54/0.207	4.90/0.084	4.61/0.187	4.43/0.282
ERIE_n	24.07/0.632	15.14/0.279	11.76/0.169	23.82/0.721	13.52/0.152	8.7535/0.040	15.11/0.206	9.06/0.044	7.74/0.024	7.39/0.019
FlowCrypt_p	11.03/0.098	10.93/0.087	10.90/0.083	11.04/0.074	10.63/0.050	10.27/0.038	10.63/0.064	10.48/0.051	10.39/0.047	10.23/0.046
FlowCrypt_n	**25.93/0.796**	**21.72/0.644**	**19.92/0.564**	**31.82/0.946**	**24.36/0.788**	**21.98/0.652**	**24.90/0.797**	**23.51/0.695**	**22.44/0.626**	**21.61/0.579**

Note: ERIE/FlowCrypt_p denotes the plain model trained without noise layer, ERIE/FlowCrypt_n denotes the model trained with noise layer.

omitted because its NPCR and UACI values are consistently close to zero, indicating its architectural limitations in diffusion capability. As shown in Table IV, FlowCrypt passes almost all NPCR and UACI tests, with only one NPCR score at the 1024×1024 scale failing the test. Although ERIE generally achieves high NPCR scores, several of its UACI values fail the corresponding statistical tests, especially at larger scales, indicating insufficient average change intensity between ciphertext pairs. These results confirm the effectiveness of FlowCrypt against differential attacks.

D. Robustness Analysis

During cloud transmission and storage, images are often subject to common degradations such as JPEG compression, Gaussian noise, Gaussian blur, median blur, cropout, dropout and salt-and-pepper (SP) Noise. To enhance robustness against these distortions, FlowCrypt is trained using a differentiable noise layer inspired by MBRS [55]. Non-differentiable distortions are implemented using differentiable surrogates (e.g., JPEGSS [56] for JPEG compression and the Kornia differentiable median-blur operator [57]). For discrete masking-based distortions (Cropout/Dropout/SP noise), we adopt a stochastic masking strategy similar to a Straight-Through Estimator (STE), where the random sampling process is treated as a constant during the backward pass, allowing gradients to propagate directly through active pixel values. Parameter distributions used for each distortion are summarized in Table V.

The recovery performance under these seven distortions is summarized in Table V, and visual comparison results are shown in Fig. 11. For a fair comparison, we report both plain variants trained without the noise layer (denoted as ERIE_p and FlowCrypt_p) and noise-layer-trained variants (denoted as ERIE_n and FlowCrypt_n) for the two end-to-end learnable methods, while other baselines are evaluated in their default configurations due to the lack of a comparable retraining setup. Overall, FlowCrypt achieves the best or near-best recovery quality across cropout, dropout, noise, blur, and JPEG distortions. Under SP noise with a ratio of 0.05, FlowCrypt maintains a recovery quality of 23.09 dB, while under Gaussian noise with a ratio

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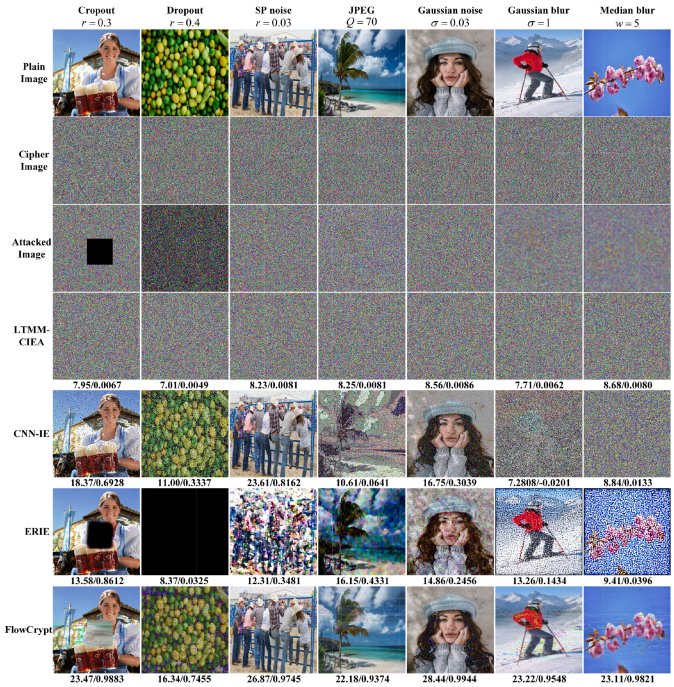

Fig. 11. Examples of decryption results against common distortions for different methods. The data under decryption results represent the PNSR and SSIM of decrypted image with respect to plain image.

TABLE VI
BENCHMARK COMPARISONS OF COMPUTATION COMPLEXITY

Method	Params	FLOPs	Enc Time(s)	Dec Time(s)	Total Time(s)
LTMM-CIEA	-	-	2.988	2.945	5.933
CNN-IE	-	-	3.951	3.983	7.934
ERIE	2.27M	37.27G	0.024	0.022	0.046
FlowCrypt	**1.01M**	**33.28G**	**0.013**	**0.011**	**0.024**

of 0.05, it outperforms competing methods by 15.65 dB. For blur distortions, FlowCrypt reaches 31.82 dB under Gaussian blur and remains the top performer under strong median blur (7×7), exceeding others by 14.32 dB. For JPEG compression (quality factor 50-90), FlowCrypt consistently yields the highest robustness. Notably, even under the non-robust setting, FlowCrypt$_p$ consistently outperforms ERIE$_p$ across all seven distortions, suggesting stronger empirical robustness of the proposed architecture before applying robustness-oriented augmentation. These results highlight the effectiveness of the invertible network in interaction between encryption and decryption, contributing to robustness against a wide range of real-world distortions.

E. Complexity Analysis

In cloud privacy scenarios, lightweight model size and computational efficiency are critical for deployment. We summarize the model parameters, Floating Point Operations (FLOPs) and evaluate the average inference encryption and decryption time on 1000 color images of size $N = 256 \times 256$, with results shown in Table VI. In the non-neural phase of FlowCrypt, the theoretical complexity of KeyGen(PBKDF), Random Split and combine is $O(N)$, where N denotes the image size, because the key stream generation and spatial reindexing operations are proportional to the image size. The mask generation exhibits a slightly higher complexity of $O(NlogN)$ due to the sorting operation. Therefore, the maximum theoretical bound for the non-neural phase is $O(NlogN)$. In the encryption and decryption network, FLOPs (Floating Point Operations) are utilized to quantify the computational complexity. For a convolution layer with output spatial size $H \times W$, kernel size $k \times k$, and channel dimensions C_{in} and C_{out}, the FLOPs are approximated as

$$\text{FLOPs}_{\text{conv}} = 2 \cdot H \cdot W \cdot C_{\text{in}} \cdot C_{\text{out}} \cdot k^2,$$

where the factor 2 accounts for multiply-accumulate operations. The FLOPs of a single DenseNet Block (DB) is computed by summing the FLOPs of its constituent convolutional layers: $\text{FLOPs}_{\text{DB}} = \sum_{l=1}^{L} \text{FLOPs}_{\text{conv}}^{(l)}$. FlowCrypt employs 4 key-conditioned affine coupling blocks, each containing 4 DBs, totaling 16 DB units. Benefiting from the random split operation, the network processes feature maps with a reduced spatial resolution of $H \times W/2$ while maintaining a minimal channel dimension ($C_{in} = 4, C_{out} = 3$). Based on the architectural parameters, the total computational cost of FlowCrypt is calculated as approximately 33.28 GFLOPs. In contrast, ERIE adopts a deeper architecture of 9 coupling blocks with 3 DBs each, totaling 27 DB units. Although the Discrete Wavelet Transform (DWT) reduces the spatial resolution to $1/4$ ($H/2 \times W/2$), it simultaneously expands the channel dimension from 3 to 12. Since the complexity of convolutional layers scales quadratically with the number of channels, the increased channel density per pixel negates the gains from spatial downsampling. Consequently, each DB in ERIE is computationally denser than those in FlowCrypt, leading to higher overall FLOPs and slower inference time than those in FlowCrypt in practice, which is consistent with the empirical measurements. Compared to other methods, FlowCrypt achieves the smallest model size and the highest runtime efficiency, contributes to the lightweight deployment in edge devices.

F. Ablation Study

1) Architecture of ACBs: To present the effectiveness of the proposed conditional ACB, we compare its encryption effect with that of RB architecture. As shown in Table VII, the key conditioned ACBs exhibit a more secure encryption effect, with entropy increasing by 1.07 compared to RB. Moreover, the recovery result of CACB also outperforms RB by 47.2 dB, further validating the superior performance of CACB in image encryption tasks.

2) Split Strategy: As shown in Table VII, split strategy plays a critical role in improving the encryption effect of FlowCrypt. The objective of split strategy is to partition the input into two equal-shape and complementary parts to match the dual-input affine coupling design in flow models. Following early coupling-based flows [37], we adopt the chessboard split as a deterministic baseline. Channel-wise split is not adopted since a 3-channel input leads to imbalanced partitions. For encryption, we further employ random split to introduce randomness before encryption

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TABLE VII
ABLATION STUDY OF THE ACB ARCHITECTURE, SPLIT STRATEGY, CIPHER LOSS, SECRET MAP AND TRIPLET LOSS

Module	Recovery Quality		Encryption Effect	
	PSNR(dB)↑	SSIM↑	Entropy	Correlation↓
(1) ACB architecture				
RB	68.86	0.9998	6.9186	0.0022
Conditional ACB	115.76	0.9999	7.9934	0.0021
(2) Split strategy				
Chessboard	80.07	0.9999	7.9552	0.2476
Random	99.99	0.9999	7.9780	0.0968
Random + PBKDF	115.76	0.9999	7.9934	0.0021
(3) Cipher Loss				
w/o $\mathcal{L}_{uniform}$	119.69	0.9999	4.0912	0.0183
w/o $\mathcal{L}_{corr}$	95.53	0.9999	7.9922	0.0266
w/ All	115.76	0.9999	7.9934	0.0021
(4) Secret map				
w/o Map K	88.89	0.9999	7.9667	0.0746
w/ Map K	115.76	0.9999	7.9934	0.0021

Module	Authorized Recovery		Unauthorized Recovery	
	PSNR(dB)↑	SSIM↑	PSNR(dB)↓	SSIM↓
(5) Triplet Loss				
w/o Triplet Loss	99.41	0.9999	11.09	0.6681
w/ Triplet Loss	115.76	0.9999	9.19	0.5521

Note: Entropy is computed on 8-bit images (ideal value is 8). Authorized recovery is evaluated by PSNR/SSIM (higher is better) and unauthorized recovery under a wrong key is evaluated by PSNR/SSIM (lower is better).

model. Specifically, we compare three strategies: 1) chessboard split, which partitions the image in a regular grid; 2) random split using a fixed seed; and 3) random split with PBKDF, where a user password is transformed into a key. From the first to second row, entropy increases by 0.0228, and pixel correlation decreases by 0.1508, indicating the effectiveness of the random split strategy. The PBKDF-based split further improves histogram uniformity, and both entropy and pixel correlation approach ideal values, contributing to more secure key generation and cipher images.

3) Importance of Cipher Loss and Triplet Loss: To promote desirable statistical properties of cipher images, we introduce two cipher regularizers: 1) a uniform loss $\mathcal{L}_{uniform}$ implemented as a KL divergence that encourages the ciphertext distribution to approach a uniform distribution, and 2) an adjacency correlation loss $\mathcal{L}_{corr}$ that penalizes correlations between adjacent pixels. Table VII shows that removing $\mathcal{L}_{uniform}$ markedly weakens the encryption effect, leading to a sharp entropy drop and increased correlation, which indicates that the KL-based uniformity regularization is critical for mitigating statistical leakage. Removing $\mathcal{L}_{corr}$ yields noticeably higher adjacent-pixel correlation while keeping entropy nearly unchanged, suggesting that it primarily suppresses local structural dependencies. When both losses are applied, FlowCrypt achieves ciphertext statistics close to the ideal entropy (8-bit) with near-zero correlation, validating the necessity of jointly enforcing global uniformity and local decorrelation. We further employ the triplet loss $L_{\text{TriRecover}}$ to improve key sensitivity by simultaneously encouraging accurate recovery under the correct key and discouraging plausible reconstructions under incorrect keys. As reported in Table VII, $L_{\text{TriRecover}}$ reduces the unauthorized (wrong-key) recovery quality while maintaining high authorized recovery quality, thereby strengthening resistance to incorrect-key decryption.

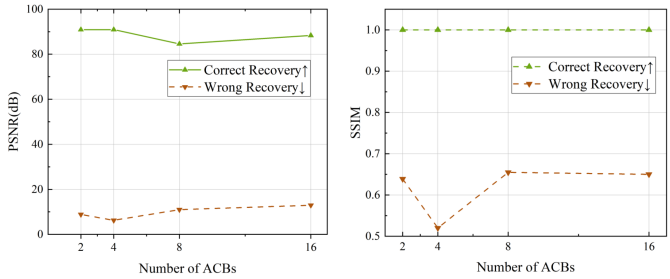

Fig. 12. Recovery performance on COCO measured by PSNR and SSIM versus the number of ACBs.

4) Number of ACB Blocks: To simultaneously achieve high-fidelity reconstruction, uniformly distributed cipher images, and a lightweight design, we conduct experiments with different values of the hyper-parameter N (i.e., the number of ACBs), as shown in Fig. 12. When $N = 4$, FlowCrypt achieves optimal recovery performance while maintaining a relatively lightweight model size.

5) Importance of Map K: The secret map K generated from the secret key k, is designed to enhance the encryption effect by introducing key-dependent randomness. This map is further concatenated with image features during the encryption process to strengthen security. As shown in Table VII, the incorporation of map K significantly improves the recovery quality, while increases the information entropy, and reduces the correlation between adjacent pixels in the cipher images. These results demonstrate its effectiveness in strengthening both reconstruction quality and confusion capabilities of the encryption scheme.

6) Importance of Noise Layer: The differentiable noise layer (denoted as FlowCrypt$_n$ in Table VI) is introduced to enhance FlowCrypt's robustness against common distortions. As shown in the last two rows of Table VI, incorporating the noise layer through joint training significantly enhances the distortion resilience of FlowCrypt.

G. Limitation and Discussion

Recent image encryption research has progressed from DNN-based schemes to INN-based designs, leveraging the non-linearity, end-to-end optimization and bidirectional mapping capabilities of neural networks to achieve image encryption. Building on this progression, FlowCrypt substantially advances reconstruction quality, achieving PSNR of 115 dB for decrypted images, surpassing the first proposed INN-based method by about 63 dB. This gain arises from the inherently invertible flow-based architecture, along with a novel CACB design suitable for image encryption tasks and the random splitting strategy that enables an information-preserving, no-reference image design.

Related studies have explored flow-based invertible models for data hiding and watermarking [21], [22], [58], leveraging the invertible nature of normalizing flows to embed and recover information with high reconstruction quality. FlowCrypt shares this invertible paradigm, but differs fundamentally in security objective and threat model. Specifically, steganography

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and watermarking aim to generate perceptually interpretable protected images, with emphasis on hidden message extraction, ownership authentication, and robustness against common distortions [59], [60], [61], whereas deepfake detection focuses on forensic analysis of synthetic or manipulated visual content [62]. In contrast, FlowCrypt targets confidentiality by generating an unintelligible cipher image that can only be recovered with the correct cryptographic key. To this end, we retain the flow-based architecture and affine coupling principle for stable invertibility, while tailoring the design for encryption through: 1) key-conditioned random splitting, which makes the transformation structurally dependent on the secret key; 2) key-conditioned affine coupling blocks for information-preserving feature fusion rather than semantic feature extraction; and 3) Cipher-randomness objectives that encourage statistical randomness and visual unintelligibility, instead of robustness-oriented recovery of embedded information.

Despite these advancements, FlowCrypt does not yet achieve strictly lossless reconstruction. This limitation may be attributed to minor numerical inaccuracies introduced by floating-point operations during computation, which can result in slight deviations in pixel values between the original and decrypted images. Although imperceptible to human vision, these deviations prevent FlowCrypt from being classified as a fully lossless encryption scheme. In future work, we aim to explore precision-preserving techniques to eliminate such losses and enable fully lossless image recovery.

VI. Conclusion

To address the limitations of existing DL-based encryption schemes in recovery fidelity, diversified robustness, ciphertext security and practicality, this paper proposes FlowCrypt, a novel flow-based encryption framework for cloud image storage. A key-conditioned random split strategy partitions plaintext into complementary representations, injecting key-dependent randomness without auxiliary references, thereby elevating ciphertext security and enabling near-lossless recovery. The parameter-shared invertible architecture, anchored by novel conditional affine coupling blocks with KL-divergence and spatial correlation regularization, ensures bijective confusion of dual inputs and high-fidelity decryption. Extensive experiments show that FlowCrypt achieves 115 dB PSNR on four datasets, generates uniformly distributed cipher images, and maintains a compact model with only 1 M parameters, making it suitable for mobile and edge-device deployments.

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